MILITARY SPECIFICATION

ACCELEROMETER, INTEGRATING, ATOMIC WEAPON: XM12

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of accelerometer entitled Accelerometer, Integrating, Atomic Weapon: XM12, which is used with the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-48733	- Switch, Accelerometer, Atomic Weapon: XM49.
MIL-R-11470	- Radiographic Inspection; Qualification of Equipment, Operators and Procedure.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

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DRAWINGS

ORDNANCE CORPS

8839227 - Accelerometer, Integrating, Atomic Weapon: XM12.
ET-8839227 - Equipment Tabulation.

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The accelerometer shall comply with all requirements specified on the drawing (dwg) 8839227, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

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3.6 Circuit resistance.

3.6.1 Safe condition. The resistance between terminals of the accelerometer connector shall not exceed the requirements of Table I when measured as specified in 4.4.1.

Table I

Resistance Between Terminals	Specified Resistance (Ohms)	Test Code No.
1 and 8	0.300 max.	031001
2 and 9	0.025 max.	031002
2 and 11	0.025 max.	031003
12 and 14	0.025 max.	031004
12 and 15	0.025 max.	031005

3.6.2 Operated conditions resistance. The resistance between the accelerometer connector terminals shall not exceed the requirements of Table II when measured as specified in 4.4.1.

Table II

Resistance Between Terminals	Specified Resistance (Ohms)	Test Code No.
9 and 5	0.045 max.	032001
9 and 12	0.045 max.	032002
9 and 14	0.045 max.	032003
9 and 15	0.045 max.	032004
2 and 12	0.045 max.	032005
2 and 14	0.045 max.	032006
2 and 15	0.045 max.	032007
11 and 12	0.045 max.	032008
11 and 14	0.045 max.	032009
11 and 15	0.045 max.	032010

3.7 Insulation resistance.

3.7.1 Safe condition. The insulation resistance between each test point designated in Table III and all other test points in Table III shall not be less than 100×10^6 ohms when tested as a potential of 500 $\pm$ 50 volts direct current (vdc) as specified in 4.4.2.

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Table III

Test Points

Terminal 1
Terminal 2
Terminal 5
Terminal 12

Terminals 3, 4, 6, 7, 10, 13 and case (may be connected in common during test).

3.7.2 Operated condition. The insulation resistance between each test point designated in Table IV and all other test points in Table IV shall be not less than 100 x 10^6 ohms when tested at a potential of 500 ± 50 vdc as specified in 4.4.2.

Table IV

Test Points

Terminal 1
Terminal 2
Terminal 8

Terminals 3, 4, 6, 7, 10, 13 and case (may be connected in common during test).

3.8 Current carrying capability. Each contact, except terminals 1 and 8, while the accelerometer is in the operated condition, shall carry the load current specified in Figure 2. The maximum voltage drop between input terminals 2, 9 and 11, in parallel, and output terminals 12, 14 and 15, in parallel, shall not exceed 2.25 volts during the 22 second load cycle. Testing shall be as specified in 4.4.3.

3.9 Reset capability. The accelerometer shall be capable of being reset from the operated to the safe condition by pressing the flexible diaphragm on each of the four acceleration switches as specified in 4.4.4.

3.10 Environmental testing (see 4.4.5).

3.10.1 Acceleration at 80 + 20°F. The accelerometer and acceleration field shall be oriented in the direction required for operation as shown in Figure 1.

3.10.1.1 Safe condition acceleration. Electrical continuity between terminals 1 and 8 shall not be broken when the accelerometer is subjected to an acceleration of 6.0 ± 0.5 gravity units (g's) applied, for not less than 8 seconds. Testing shall be as specified in 4.4.5.1.1.

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3.10.1.2 No load operation acceleration. The accelerometer shall be subjected to the acceleration levels specified in Table V. Electrical continuity between terminals 1 and 8 shall be broken within the time limits specified in Table V. Electrical continuity between input terminals 2, 9 or 11 and output terminals 12, 14 or 15 shall be established within the time limits specified in Table V. Testing shall be as specified in 4.4.5.1.2. After the acceleration field has been removed, the accelerometer shall remain in the operated condition and shall comply with the requirements of 3.6.2 and 3.7.2.

Table V

Test No.	Acceleration Level (g's) 1 Run Each Level	Time to Operate (Seconds) Min.	Time to Operate (Seconds) Max.	Monitor Opening Time Max. (Seconds)	Test Code No.
1	13.0 ± 0.3	----	3.5	0.35	052001
2	15.0 ± 0.4	2.06	2.58	0.26	052002
3	38.0 ± 0.8	1.25	1.62	0.16	052003

3.10.1.3 Current load operation. During the operating acceleration, the contacts between the accelerometer input terminals 2, 9 and 11 in parallel, and output terminals 12, 14 and 15, in parallel, shall close and then carry a 20 ± 1 ampere resistive dc load for 60 ± 40 milliseconds. Immediately following this test, the current shall be reduced to 5.5 ± 0.3 ampere resistive dc load for a period of 22 seconds minimum. During this test, the parallel resistance shall not exceed 0.045 ohms subsequent to a 0.05 second period after closure of the first contact. Within 0.42 seconds after initial completion of circuit, the resistance shall not exceed 0.035 ohms. Testing shall be at 15.0 ± 0.4 g's and as specified in 4.4.5.1.3.

3.10.2 Operating requirements at -40°F.

3.10.2.1 Safe condition. The accelerometer shall meet the requirements of 3.10.1.1 at -40 + 5 degrees Fahrenheit (°F). Testing shall be as specified in 4.4.5.2.1.

3.10.2.2 No loading operation. The accelerometer shall meet the requirements of 3.10.1.2 when subjected to test number 2 of Table V at -40 + 5°F. Testing shall be as specified in 4.4.5.2.2. After removal of the acceleration field, the accelerometer shall comply with the requirements of 3.11 at 80 ± 20°F.

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3.10.3 Operating requirements at 160°F.

3.10.3.1 Safe condition. The accelerometer shall meet the requirements of 3.10.1.1 when tested at 160 ± 5°F. Testing shall be as specified in 4.4.5.3.1.1.

3.10.3.2 No load operation. The accelerometer shall meet the requirements of 3.10.1.2 when subjected to test Number 2 of Table V at 160 ± 5°F. Testing shall be as specified in 4.4.5.3.2. After removal of the acceleration field, the accelerometer shall comply with the requirements of 3.11 at 80 ± 20°F.

3.10.4 Impact shock.

3.10.4.1 Safe condition. The accelerometer shall withstand 40 impact shocks in the direction required to induce operation (see Figure 1) without evidence of damage affecting function. Shocks of 22 ± 2 g's peak amplitude shall be applied for 10 ± 2 milliseconds (ms) at the 10 g level. After this test, and prior to removal from the shock producing device, the circuit between terminal 5 and terminals 2, 9, or 11, and the circuit between terminal 5 and terminals 12, 14 or 15 shall show no evidence of closure, and the accelerometer shall show no evidence of partial operation. After testing, the accelerometer shall be in the safe condition and shall comply with the requirements of 3.6.1 and 3.7.1. Testing shall be as specified in 4.4.5.4.1.

3.10.4.2 Operated condition. The accelerometer shall withstand three (3) impact shocks, in the direction opposite to that required for operation, of 80 ± 8 g's peak amplitude for a duration of 3 ± 1 milliseconds (ms) at base amplitude. During the 5 ms interval immediately following the initiation of the shock pulse, the continuity between terminals 2, 9 and 11, and between terminals 12, 14 and 15 shall not be interrupted for more than 20 microseconds. Testing shall be as specified in 4.4.5.4.2.

3.10.5 Vibration.

3.10.5.1 Safe condition. The accelerometer shall withstand vibration in each of the three mutually perpendicular axes at the frequencies, amplitudes, and for the times specified in Table VI without evidence of damage or functional deterioration. During vibration, the circuits between terminals 2, 9 and 11 and between 12, 14 and 15 shall show no evidence of closure. After completion of vibration the accelerometer shall be in the safe condition and shall comply with the requirements of 3.6.1 and 3.7.1. Testing shall be as specified in 4.4.5.5.1.

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Table VI

Frequency Range Hz	Double Displacement or Peak G Level (g's)	Cycle Time To Cover Frequency Range (Minutes)	Test Time For Each Axis In Minutes
10-20-10	2 ± 0.1 g level	3 ± 0.5	10 ± 0.1
20-50-20	0.036 ± 0.0018 inch	3 ± 0.5	10 ± 0.1
50-500-50	5 ± 0.25 g level	3 ± 0.5	10 ± 0.1

3.10.5.2 Operated condition. The accelerometer shall withstand flight vibration through a frequency range of 75-1000-75 Hertz (Hz) at a peak level of 5 ± 0.25 g's for 3.0 ± 0.3 minutes per axis, followed by a vibration of 450 Hz at a peak of 5 ± 0.25 g's for 2.0 ± 0.2 minutes per axis without evidence of damage or functional deterioration. During vibration the electrical resistance between input terminals 2, 9 and 11 and between output terminals 12, 14 and 15 shall not exceed 0.045 ohms and the circuit continuity shall not be interrupted for more than 20 microseconds. After completion of vibration, the accelerometer shall be in the operated condition and shall comply with the requirements of 3.6.2 and 3.7.2. Testing shall be as specified in 4.4.5.5.2.

3.11 Radiographic examination. Actuation of all four acceleration switches shall be indicated when tested as specified in 4.4.6. Equipment and operators shall be qualified in accordance with MIL-R-11470.

3.12 Acceleration switch. Records shall be furnished showing that the acceleration switches employed have been inspected and accepted in accordance with the requirements of MIL-A-48733(PA).

3.13 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.14 Workmanship.

3.14.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.14.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.14.3 Burrs. Parts shall be free from burrs.

3.14.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.14.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 6 complete assemblies (see dwg 8839227) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table VII. Tests shall be conducted in order specified in Table VII.

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Table VII

First article inspection and tests

Inspection or Test	Unit Number 1	2	3	4	5	6	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)	X	X	X	X	X	X		X
2. Circuit resistance (safe) (see 3.6.1)	X	X	X	X	X	X		X
3. Insulation resistance (safe) (see 3.7.1)	X	X	X	X	X	X		X
4. Acceleration at 80°F (safe) (see 3.10.1.1)	X	X	X	X	X	X		X
5. Acceleration at 80°F (no load) (see 3.10.1.2)	X	X	X	X	X	X		X
6. Circuit resistance (operated) (see 3.6.2)	X	X	X	X	X	X		X
7. Insulation resistance (see 3.7.2)	X	X	X	X	X	X		X
8. Reset capability (see 3.9)	X	X	X	X	X	X		X
9. Operation requirement at -40°F (safe) (see 3.10.2.1)			X	X				X
10. Operation requirement at -40°F (operated) (see 3.10.2.2)			X	X				X
11. Operation requirement at 160°F (safe) (see 3.10.3.1)					X	X		X
12. Operation requirement at 160°F (operated) (see 3.10.3.2)					X	X		X
13. Impact shock (safe) (see 3.10.4.1)	X	X					X	
14. Vibration (safe) (see 3.10.5.1)		X	X		X	X	X	
15. Acceleration at 80°F (no load) (see 3.10.1.2)	X	X	X	X	X	X		X
16. Vibration (operated) (see 3.10.5.2)		X	X		X	X	X	
17. Current carrying capability (see 3.8)	X	X	X	X	X	X		X
18. Reset capability (see 3.9)	X	X	X	X	X			X
19. Current load operation (see 3.10.1.3)	X	X		X	X			X
20. Impact shock (operated) (see 3.10.4.2)	X			X		X	X	

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table VIII and Table IX), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table VIII
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table IX
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Accelerometer before potting (see dwg 8839227)

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Improper soldering (see 3.13)	Visual
Minor:	None defined.	

4.3.2.1.2 Accelerometer after potting (see dwg 8839227).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Improper or missing component	Visual
	102. Damaged or loose component	Visual
	103. Improper assembly or component	Visual
	104. Improper overall height	Gage
	105. Improper mounting hole location (3 holes)	Gage
	106. Wires protruding from potting (Wires may be visible but not protrude)	Visual
	107. Evidence of poor workmanship (see 3.14)	Visual
	108. Objective evidence of relating to material missing or improper (see 3.1)	Visual

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Minor:
201. Marking improper, misleading, illegible or missing (see dwg 8839227).....Visual

4.3.2.1.3 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packing, and packaging not in accordance with MIL-STD-1169	Visual
102.	Marking not in accordance with MIL-STD-1169	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.14)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Circuit resistance (safe)	Major
b. Insulation resistance (safe)	Major
c. Safe condition acceleration	Major
d. No load operation acceleration	Major
e. Circuit resistance (operated)	Major
f. Insulation resistance (operated)	Major
g. Reset capability	Major
h. Impact shock (safe)	Major
i. Vibration (safe)	Major
j. Vibration (operated)	Major

Of these tests, h, i and j are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

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4.3.3.1 Circuit resistance (safe), circuit resistance (operated), insulation resistance (safe), insulation resistance (operated), safe condition acceleration and reset capability. Each accelerometer in the lot shall be tested in accordance with 4.4.1, 4.4.2, 4.4.5.1.1 and 4.4.4. If any item fails to comply with 3.6.1, 3.6.2, 3.7.1, 3.10.1.1 and 3.9, the item shall be rejected and removed from the lot.

4.3.3.2 No load operation acceleration. Each accelerometer in the lot shall be subjected to these tests in sequence (see 4.4.5.1.1) by groups in accordance with the following schedule:

Group I (30 percent of lot). The accelerometers shall be tested in accordance with 3.10.2.2 for compliance with Test 2, Table V.
Group II (30 percent of lot). The accelerometers shall be tested in accordance with 3.10.1.2 for compliance with Test 2, Table V.
Group III (20 percent of lot). The accelerometers shall be tested in accordance with 3.10.3.2 for compliance with Test 2, Table V.
Group IV (20 percent of lot). The accelerometers shall be tested in accordance with 3.10.1.2 for compliance with Test 3, Table V.

If a failure occurs in any group, reject the item and continue to test the remaining quantity for that particular group, and then follow below listed schedule of testing:

4.3.3.2.1 Group I Failure. If a failure occurs in Group I, subject the balance of 70% of the lot to the following tests:

(a) 20% of the lot shall be tested in accordance with the requirements of Group I.

(b) 25% of the lot shall be tested in accordance with the requirements of Group II.

(c) 25% of the lot shall be tested in accordance with the requirements of Group III.

(d) All of the above (70%) shall be tested in accordance with the requirements of Group IV.

4.3.3.2.2 Group II Failure. If a failure occurs in Group II, subject the balance of 40% of the lot to the following test:

(a) All of the above (40%) shall be tested in accordance with the requirements of Group II.

(b) 30% of the units from Group I, II and III shall be tested in accordance with the requirements of Group IV.

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4.3.3.2.3 Group III Failure. If a failure occurs in Group III, subject the balance of 20% of the lot to the following test:

(a) All of the above (20%) shall be tested in accordance with the requirements of Group III.

(b) 30% of the units from Group I and II shall be tested in accordance with the requirements of Group IV.

4.3.3.2.4 Group IV Failure. If a failure occurs in Group IV, subject an additional quantity of accelerometers to the following test:

(a) 30% of the units from Groups I and II shall be tested in accordance with the requirements of Group II.

4.3.3.3 Impact shock (safe), vibration (safe) and vibration (operated). Two sample accelerometers shall be selected at random for each test indicated above from each lot of 100 accelerometers, or portion thereof produced, (total of six samples). The samples shall be tested in accordance with 4.4.5.4.1, 4.4.5.5.1 and 4.4.5.5.2. Accelerometers subjected to shock (safe) and vibration (safe) tests shall, upon completion comply with the requirements of 3.10.1.2, Test 2 of Table V when subjected to two acceleration runs as specified in 4.4.5.1.2. If any item fails to comply with 3.10.4.1.1, 3.10.5.1 and 3.10.5.2, or 3.10.1.2 Test 2 of Table V where applicable, the lot represented by the sample in which a failure occured shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). ELs for assemblies are listed on the Equipment Tabulation, ET-8839227. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any devistion from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part of Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

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4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit resistance. The circuit resistance shall be measured by means of a suitable low electrical resistance measuring instrument which has been connected to the terminals specified in 3.6. A current of 100 milliamperes (ma) maximum, shall be applied in conducting this test.

NOTE: Care must be exercised to avoid bending of terminals pins and breakage of glass seals.

4.4.2 Insulation resistance. Insulation resistance shall be measured by means of an instrument capable of measuring megohms and by applying a current not to exceed 5 milliamperes (ma). The potential shall be applied until meter reading becomes steady but the test shall not exceed one (1) minute in duration. The terminals of the accelerometer connector shall be connected as specified in 3.7.

4.4.3 Current carrying capability. The accelerometer terminals shall be connected to a device capable of reproducing the current-time cycle of Figure 2. The voltage drop between the input and output terminals shall be monitored during this test.

4.4.4 Reset capability. The accelerometer shall be reset from the operated to the safe condition by pressing a soft rubber instrument against the flexible diaphragm of each of the four acceleration switches. Resetting can be detected by listening for the weights to release inside the acceleration switches and check for circuit continuity between terminals 1 and 8.

CAUTION: Do not apply pressure against the flexible diaphragms with fingernails or other hard objects.

4.4.5 Environmental testing (see 3.10).

4.4.5.1 Acceleration at 80 + 20°F.

4.4.5.1.1 Safe condition acceleration. Acceleration tests shall be run on a centrifuge using a rotating, shockless fixture to bring the accelerometer into proper orientation for operation. The major axis of the accelerometer shall be placed perpendicular to the arm of the centrifuge within 2 degrees when the fixture is in its initial position. The fixture shall rotate the accelerometer into the acceleration field after the proper acceleration value is attained. At this time the accelerometer shall be oriented as shown in Figure 1. The major axis of the accelerometer shall be parallel to the direction of acceleration within 2°. A continuity tester shall be employed for this test.

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4.4.5.1.2 No load operation-acceleration. Testing shall be conducted on a centrifuge using a rotating, shockless fixture to orient the accelerometer for operation. The major axis of the device shall be perpendicular to the arm of the centrifuge within 2° when the fixture is in its initial position. The fixture shall rotate the accelerometer into the acceleration field only after the proper acceleration is attained. At that time, the accelerometer shall be oriented as shown on Figure 1. The major axis of the accelerometer shall be parallel to the direction of acceleration within 2°. The fixture shall be capable of rotating the accelerometer from initial position to final position in 0.047 to 0.150 seconds when the acceleration level is 15 g's and in 0.047 to 0.110 seconds when the acceleration level is 38 g's. Accelerometer operating times shall be measured from the time of release of the fixture from its initial position to opening of the monitoring contacts (circuit between terminals 1 and 8) and to closure of activating contacts (circuit between terminals 2, 9 or 11 and 12, 14 or 15). After the last centrifuge test of Table V, the accelerometer shall be removed from the centrifuge in the operated condition and shall meet requirements of 4.4.1 and 4.4.2.

4.4.5.1.3 Current load operation. The current load operation test shall be performed by using a centrifuge and a shockless test fixture of the type described in 4.4.5.1.2. A circuit capable of supplying the current load requirements of 3.10.1.3 shall be employed. The resistance shall be monitored during this test in accordance with 3.10.1.3.

4.4.5.2 Operating tests at -40°F.

4.4.5.2.1 Safe condition. Testing shall be as specified in 4.4.5.1.1 except as follows: The accelerometer shall be placed in an environmental test chamber and maintained at -40 ± 5°F for a minimum of four hours. The test chamber shall be stabilized at -40 ± 5°F per 6.4 and shall maintain that temperature throughout the period of use. The accelerometer shall then be rapidly transferred from the environmental test chamber to a temperature box mounted on the centrifuge. The centrifuge temperature box shall be stabilized at -40 ± 5°F per 6.4 and remain at that temperature throughout the test. The accelerometer shall remain in the temperature box for a minimum of 15 minutes prior to testing at the specified temperature.

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4.4.5.2.2 No load operation. Testing shall be conducted as specified in 4.4.5.1.2 except as follows. The accelerometer shall be placed in an environmental test chamber and maintained at -40 ± 5°F for a minimum of four hours. The test chamber shall be stabilized at -40 ± 5°F per 6.4 and shall maintain that temperature throughout its use. The accelerometer shall then be rapidly transferred from the environmental test chamber to a temperature box mounted on the centrifuge. The centrifuge temperature box shall be stabilized at -40 ± 5°F per 6.4 and maintained at that temperature throughout the test. The accelerometer shall remain in the temperature box for a minimum of 15 minutes prior to testing at the specified temperature.

4.4.5.3 Operating tests at 160°F.

4.4.5.3.1 Safe condition. Same as 4.4.5.2.1 except substitute "160 ± 5°F" for "-40 ± 5°F".

4.4.5.4 Impact shock.

4.4.5.4.1 Safe condition. The accelerometer shall be rigidly mounted to a suitable shock producing device and shall be shocked in the operating direction as shown on Figure 1. Shocks shall be applied at the rate of at least one shock per minute. The monitor contacts shall be checked for continuity after this test. Following this test, the accelerometer shall be tested as specified in 4.4.1 and 4.4.2.

4.4.5.4.2 Operated condition. The accelerometer shall be rigidly mounted to a suitable shock-producing device and shocked three times in the direction opposite to that required for operation. Each shock shall be applied for the specified time. Continuity between input and output terminals shall be monitored during this test with an instrument capable of detecting circuit openings for 20 microseconds duration or greater. Following this test, the accelerometer shall be tested as specified in 4.4.1 and 4.4.2.

4.4.5.5 Vibration.

4.4.5.5.1 Safe condition. The accelerometer shall be rigidly mounted to a fixture capable of readily supporting the accelerometer in each of three mutually perpendicular axes and the fixture shall be mounted on a suitable vibration producing apparatus. The accelerometer shall be vibrated at the frequencies and amplitudes and for the times specified in Table VI. Following completion of the test, the accelerometer shall be tested as specified in 4.4.1 and 4.4.2.

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4.4.5.5.2 Operated condition. The accelerometer in the operated condition shall be rigidly mounted to a fixture capable of being readily positioned in each of three mutually perpendicular axes. The fixture shall be mounted on a suitable vibration producing apparatus and vibrated for the times specified. The contact resistance shall be measured with a suitable resistance measuring device and continually shall be monitored during this test with an instrument capable of detecting circuit openings of 20 microseconds minimum duration. Circuit resistance and continuity may be monitored during separate test runs. Following this test, the accelerometer shall be tested as specified in 4.4.1 and 4.4.2.

4.4.6 Radiographic examination. The accelerometer shall be oriented as indicated on Figure 3 and then x-rayed. The x-ray beam shall be focused such that it is possible to determine that all acceleration switches are in the latched (operated) position.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The accelerometer shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The accelerometer shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The accelerometer shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 3.5, 3.6.1, 3.6.2, 3.10.1.2 and 4.3.4.1). Data specified in 3.5, 3.6.1, 3.6.2, 3.10.1.2 and 4.3.4.1 will be delivered as idendified on a DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

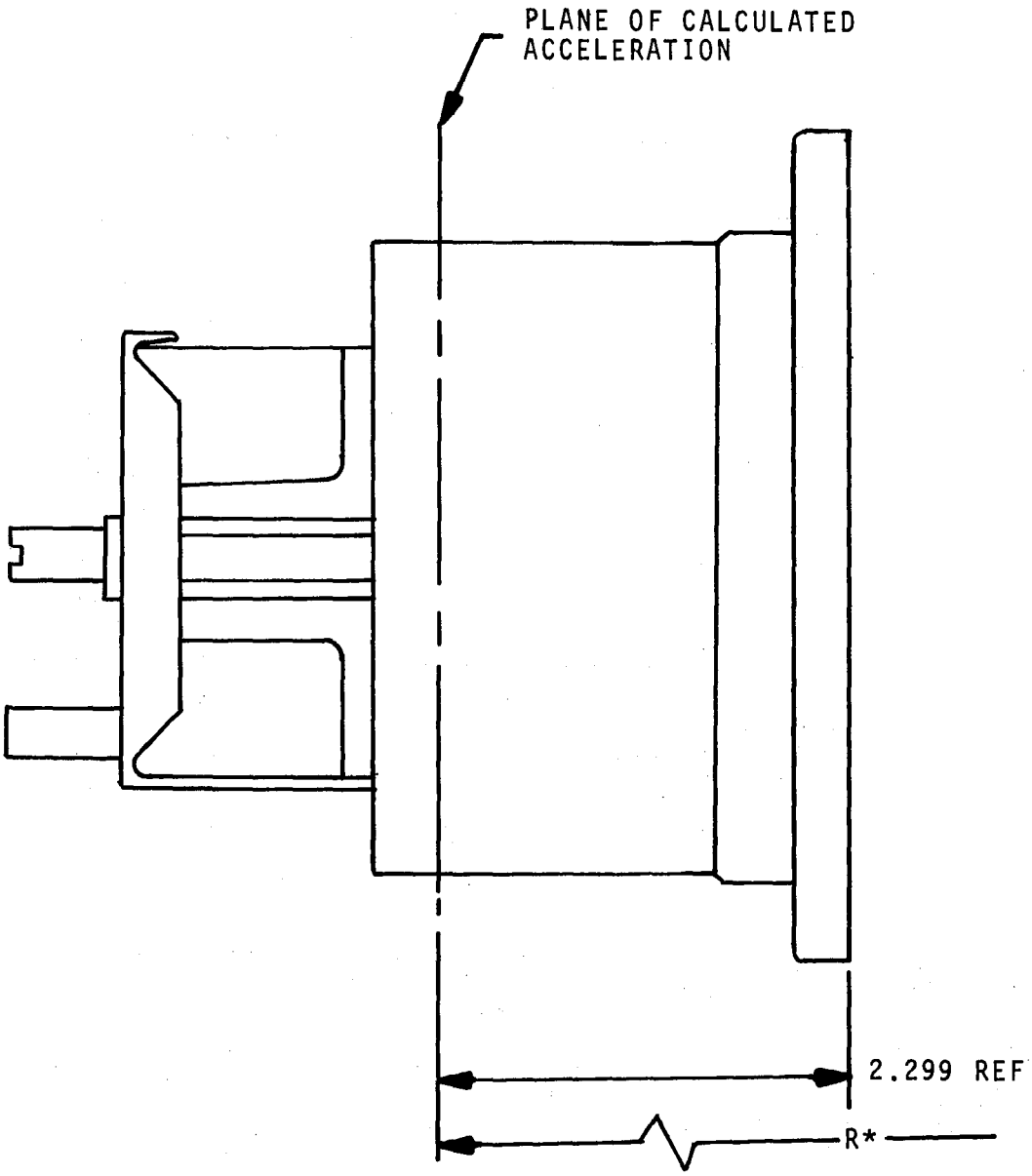
6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2318 dated 21 December 1965 and Amendment 1 dated 28 June 1967.

Custodian:
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Preparing activity:
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DIRECTION OF ROCKET ACCELERATION FOR OPERATION.

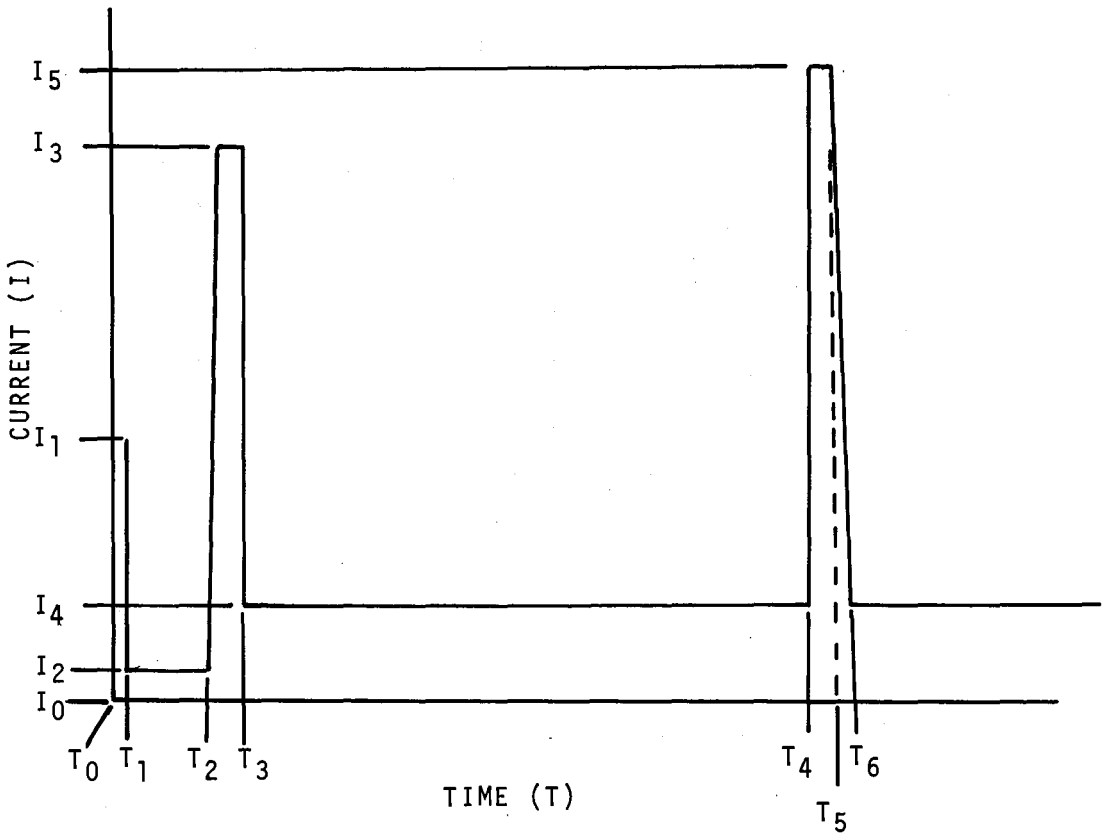
*R IS THE CENTRIFUGE ARM DISTANCE REQUIRED FOR APPROPRIATE ACCELERATION LEVEL (g's).

FOR TESTING ORIENT ACCELEROMETER SO THAT CONNECTOR POINTS TOWARD CENTER OF CENTRIFUGE.

FIGURE 1

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ACCELEROMETER LOAD CURRENT



Time

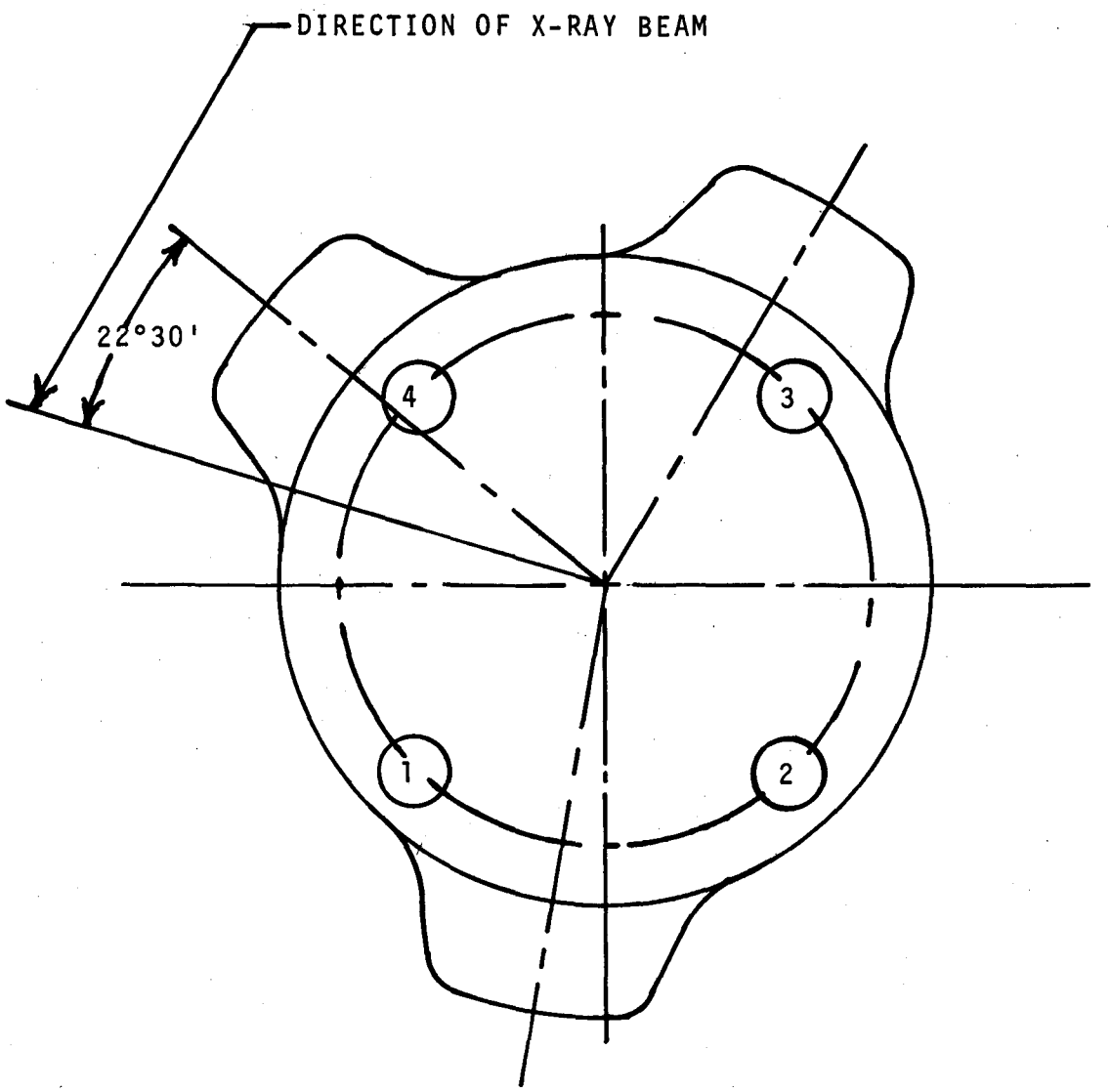
T_0 Time of load application for current carrying capability.
T_1 Minus T_0 = 60 milliseconds ± 40.0 milliseconds.
T_2 Minus T_0 = 2 seconds ± 0.5 seconds.
T_3 Minus T_2 = 125 milliseconds ± 75 milliseconds.
T_4 Minus T_0 = 21 seconds ± 1.0 seconds.
T_5 Minus T_4 = 5 milliseconds + 4.0 milliseconds -0.0 milliseconds.
T_6 Minus T_0 = 22 seconds ± 0.5 seconds.

Current-Amperes

I_0 = 0
I_1 = 14.0 ± 0.8
I_2 = 2.0 ± 0.2
I_3 = 35.0 ± 3.0
I_4 = 5.5 ± 0.3
I_5 = 45.0 ± 5.0

Figure 2

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ORIENTATION FOR X-RAY TESTING

Figure 3

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